

MD MAHFUZUR RAHMAN

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PROFESSIONAL SUMMARY

Mechanical engineer specializing in biomedical devices, with 6+ years of experience developing and validating wearable medical devices — bridging laboratory-grade clinical measurement and accessible, home-use health technology. Expertise spans the full development pipeline: microfabrication, custom analog electronics and PCB design, sensor integration, and advanced biomedical signal processing (digital filtering, FFT, wavelet transforms, EMD, HHT) for physiological signal analysis. Skilled in drafting and executing verification/validation test protocols, statistical data analysis, and IRB/regulatory-compliant human subjects research. Proven ability to coordinate multi-stakeholder projects on deadline and collaborate across engineering and clinical teams. NSF-funded researcher with 6 peer-reviewed publications.

EDUCATION

- **Ph.D., Mechanical & Aerospace Engineering** Defended; degree conferral August 2026
Old Dominion University, Norfolk, VA GPA: 3.87 / 4.0
Dissertation: Time-Frequency Analysis of Arterial Pulse Signals of Cardiovascular Disease Patients Measured by a Microfluidic Based Tactile Sensor
- **M.S., Applied Physics, Electronics & Communication Engineering** March 2012
University of Dhaka, Bangladesh GPA: 3.73 / 4.0 (WES-evaluated)
- **B.S., Applied Physics, Electronics & Communication Engineering** September 2010
University of Dhaka, Bangladesh GPA: 3.81 / 4.0 (WES-evaluated)

TECHNICAL SKILLS

- **Microfabrication:** Photolithography; thin-film deposition (sputtering); soft lithography (SU-8 molds); PDMS casting & plasma bonding; 3D-printed molds; end-to-end device fabrication
- **Instrumentation:** Analog Front-End (AFE) Design, Schematic & PCB Layout (KiCad), Circuit Simulation (Multisim), Sensor-to-DAQ Integration (NI DAQ, LabVIEW)
- **Test, Verification & Validation:** Test protocol drafting & execution; performance qualification; device characterization (sensitivity, linearity, frequency response, repeatability); technical validation reporting
- **Programming & Software:** MATLAB; Python; Simulink; LabVIEW; AutoCAD(2D); Autodesk Inventor(3D); Microsoft Office (Word, Excel, PowerPoint, Outlook); LaTeX
- **Clinical & Regulatory:** IRB protocol design & compliance; human subjects research; REDCap data management; GCP/regulatory documentation; CITI certified (Biomedical Research, active through March 2029)
- **Project & Collaboration:** Multi-stakeholder project coordination; deadline management; cross-functional teamwork; mentoring students and technicians

PROFESSIONAL EXPERIENCE

Graduate Research Assistant Sept 2019 – Present
Old Dominion University, Dept. of Mechanical & Aerospace Engineering, Norfolk, VA

- **Microfluidic Tactile Sensor:** Designed and fabricated end-to-end soft polymer (PDMS) microfluidic sensors and custom 5-channel analog readout PCBs; characterized sensitivity, linearity, and frequency response, achieving 1.5 mm spatial resolution
- **Motion Artifact Modeling:** Developed and experimentally validated an analytical model showing motion artifacts alter sensor-tissue system parameters; built MATLAB/Python signal processing pipelines (FFT, EMD, HHT, CDM, HVD) for time-frequency analysis
- **Multi-BMI Clinical Study:** Designed and executed a multi-visit IRB-approved clinical study acquiring multimodal cardiovascular data (sensor, Doppler ultrasound, tonometry, blood pressure); validated derived parameters against clinical reference standards
- **Accelerometer Feasibility Study:** Designed a custom flexible PCB with analog/digital accelerometers and signal-conditioning front-end; conducted IRB-approved feasibility study identifying key sensing limitations
- **Cross-Cutting:** Drafted/executed V&V test protocols and technical reports; managed REDCap data to regulatory/NSF compliance; coordinated multi-stakeholder studies on deadline; contributed to 6 publications; mentored students

Graduate Teaching Assistant Sept 2019 – Dec 2025
Old Dominion University, Dept. of Mechanical & Aerospace Engineering, Norfolk, VA

- Served as primary instructor for MAE 203 (Materials Science Lab); supported Dynamics, Solid Mechanics, Materials Science, and Engineering Design

- Delivered lab instruction, graded assessments, mentored student teams, and supported ABET outcome documentation

Manager — Merchandising & Production

Apr 2017 – Aug 2019

Afiya Knitwear Ltd., Dhaka, Bangladesh

- Managed end-to-end production planning, costing, quality compliance, and vendor coordination for international manufacturing contracts in a deadline-driven environment
- Led cross-functional teams ensuring on-time delivery, cost control, and adherence to quality and compliance standards

Lecturer — Electronics & Telecommunication Engineering

Oct 2014 – Mar 2017

Atish Dipankar University of Science & Technology (ADUST), Dhaka, Bangladesh

- Independently developed and taught 14 undergraduate engineering courses including instrumentation, electronics, measurement, and digital signal processing; mentored final-year student projects

Part-Time Lecturer — Computer Science & Engineering

Aug 2016 – Mar 2017

Daffodil Institute of Information Technology (DIIT), Dhaka, Bangladesh

- Taught introductory electrical engineering and microprocessor courses with integrated laboratory components

SELECTED PUBLICATIONS

- Rahman, M.M., et al. (2025). An Analytical Model of Motion Artifacts in a Measured Arterial Pulse Signal (Parts I & II). *Sensors*, 25(18). [Q2]
- Rahman, M.M., et al. (2025). Motion Artifacts At-Rest in Measured Arterial Pulse Signals. *Biosensors*, 15(9), 578. [Q1]
- Rahman, M.M., et al. (2021). Post-Exercise Response of Local Arterial Parameters Using a Microfluidic Tactile Sensor. *IEEE EMBC 2021*.
- *Full list (6 journal articles, 4 first-author; 2 conference proceedings) available on Google Scholar.*

HONORS & AWARDS

- Tiwari Endowed Graduate Scholarship, Old Dominion University (2021)
- National Science & ICT (NSICT) Fellowship, Bangladesh (2010)
- Merit-Based Scholarship, University of Dhaka (2008)

LEADERSHIP & INVOLVEMENT

- Volunteer, Virginia Medical Reserve Corps — COVID-19 vaccination clinics, community health screenings, and patient intake (2020–Present)
- Author/presenter, IEEE Engineering in Medicine & Biology Conference (EMBC 2021); published in ASME Journal of Engineering and Science in Medical Diagnostics and Therapy

REFERENCES

Available upon request. Primary academic reference: Dr. Zhili Hao, Professor, Old Dominion University (zlhao@odu.edu).